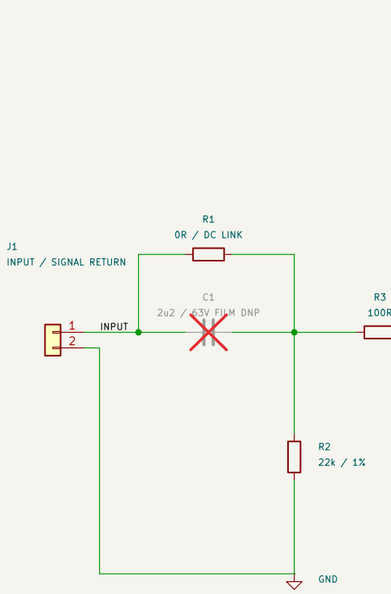
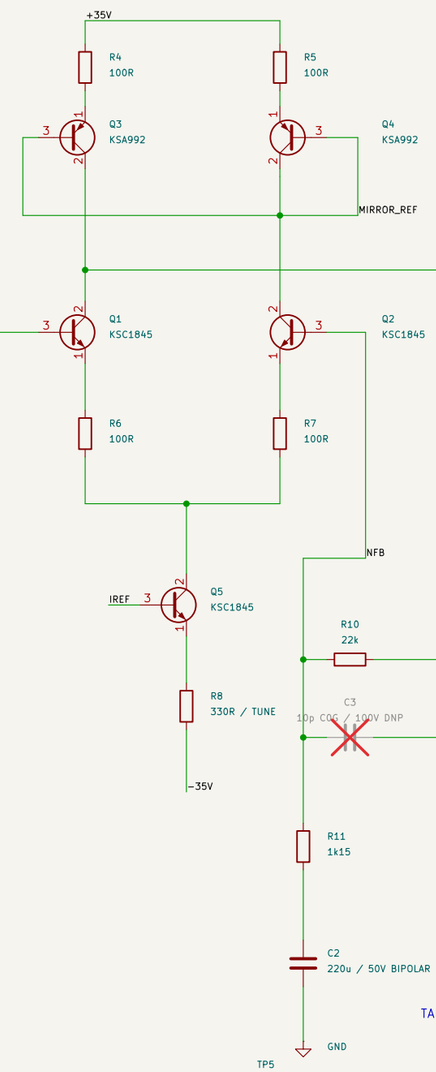
ONE MONO CHANNEL — DISCRETE CLASS AB / +/−35 V

REV A: FIRST SCHEMATIC FOR ENGINEERING REVIEW. 8 ohm target; 4 ohm capability needs SOA/thermal qualification.

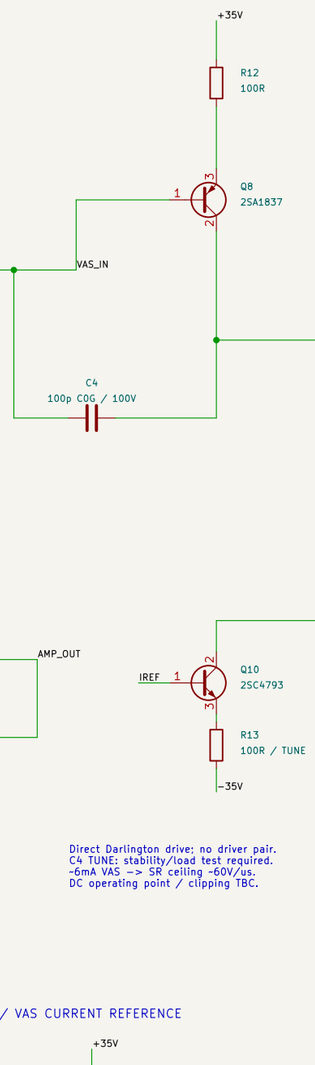
INPUT



DIFFERENTIAL / MIRROR

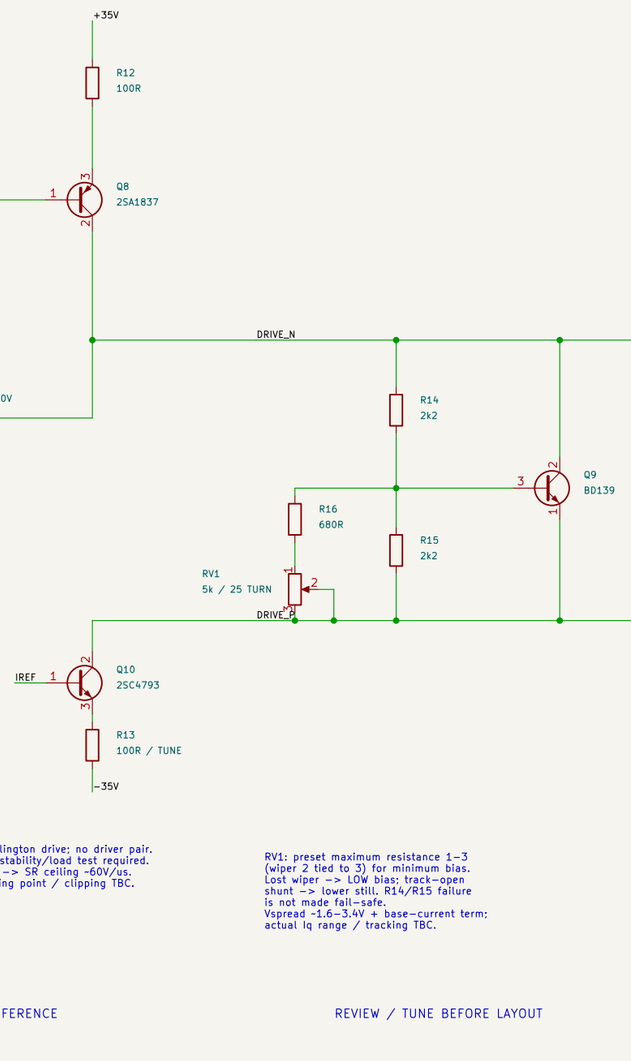


VAS / CURRENT SINK



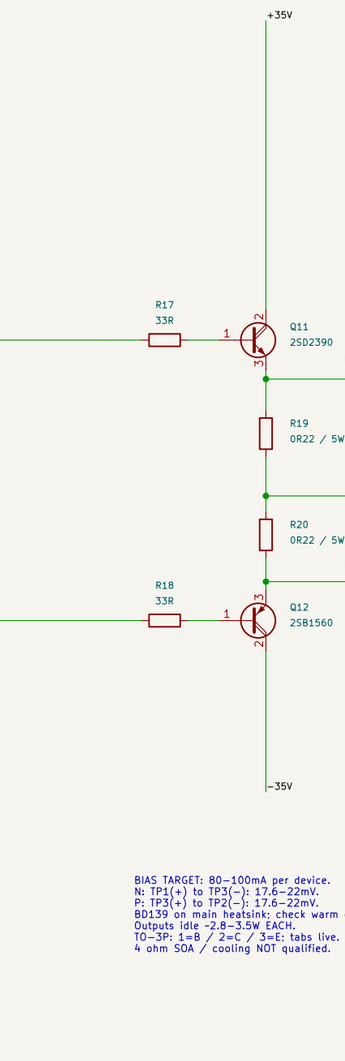
Direct Darlington drive; no driver pair.
C4 TUNE: stability/load test required.
~6mA VAS −> SR ceiling ~60V/us.
DC operating point / clipping TBC.

BIAS SPREADER



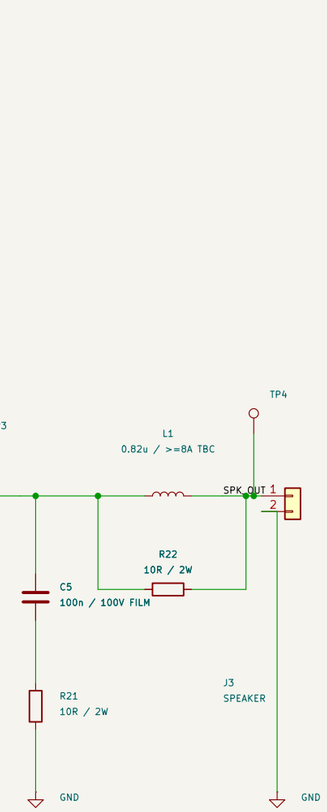
RV1: preset maximum resistance 1−3
(wiper 2 tied to 3) for minimum bias.
Lost wiper −> LOW bias; track−open
shunt −> lower still. R14/R15 failure
is not made fail−safe.
Vspread ~1.6−3.4V + base−current term;
actual Iq range / tracking TBC.

DARLINGTON OUTPUT



BIAS TARGET: 80−100mA per device.
N: TP1(+) to TP3(−): 17.6−22mV.
P: TP3(+) to TP2(−): 17.6−22mV.
BD139 on main heatsink; check warm drift.
Outputs idle ~2.8−3.5W EACH.
TO−3P: 1=B / 2=C / 3=E; tabs live.
4 ohm SOA / cooling NOT qualified.

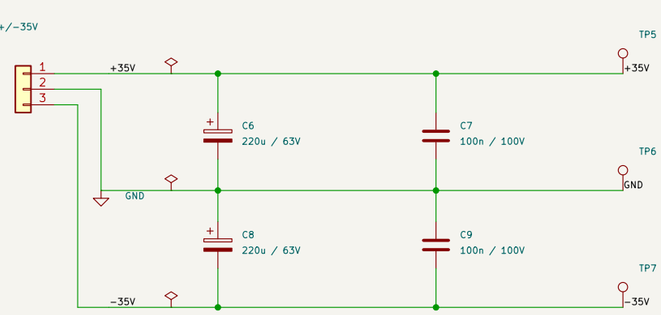
OUTPUT NETWORK



Feedback / Zobel at AMP_OUT.
Speaker connector after L1.
Speaker return / Zobel return
route separately to local star.
Nominal ~30−31Vpk −> 56−60W/8R;
rail sag / clipping must be measured.

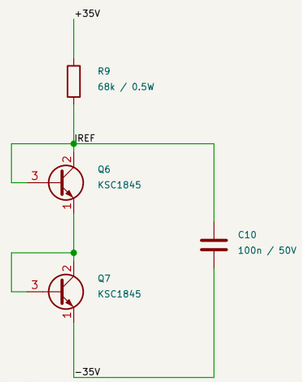
DC input link fitted. Zin ~22k.
Audio gain 20.13 (26.1dB): DC gain 1.
C2 is bipolar: no fixed DC polarity.
Q1/Q2: match and thermally pair.

EXTERNAL POWER / LOCAL DECOUPLING



One electrical GND net. Local star/reference at J2 pin 2.
Input + feedback returns form quiet branches to that point.
Speaker, Zobel and decoupling returns use separate power paths.
Generic connector and passive footprints are TBC before layout.
J2/J3: select >=10A rated terminals; no series chosen yet.

TAIL / VAS CURRENT REFERENCE



IREF ~−33.8V. Tail ~1.8mA; VAS ~6mA.
R9/R13 TUNE: diode and VBE spread matter.
Q1/Q2 collectors ~+33V, emitters ~−0.7V.
Q5 ~60mW; front−end VCEO rating 120V.
Medium−power Q8/Q10 rated 230V.

REVIEW / TUNE BEFORE LAYOUT

C4 Miller (100p start), C3 lead (DNP), R8/R13 current, RV1 bias.
Check square−wave response with 8R / 4R / capacitive loads.
Check offset, startup, clipping recovery and heatsink bias tracking.
No DC protection, current limiting or thermal protection in Rev A.
At +/−35V a single pair is NOT signed off for sustained 4R use.
Worst resistive Class−B dissipation ~15.5W/device (8R),
~31W/device (4R), plus bias and reactive−load stress.
Small−signal leadforms and all exact passive/connector packages TBC.
BD139 symbol uses onsemi ECB pin map; verify any stocked ST part
independently (ST legacy internal schematic numbering conflicts).

Schematic only. Stability, bias and 4 ohm SOA require bench review.

Sheet: /
File: DarlingtonAmp.kicad_sch

Title: DarlingtonAmp — Mono Class AB

Size: A3	Date: 2026−10−05	**Rev: A — REVIEW**
KiCad E.D.A. 10.0.5		Id: 1/1